COLE A. MURRAY

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EDUCATION

University of Illinois at Urbana-Champaign, Grainger College of Engineering GPA: 3.75/4.00
Bachelor of Science in Computer Engineering, Hoeft Technology & Management Minor Expected Graduation May 2029

Related Coursework:

Digital Systems Laboratory (FPGA), Computer Systems & Programming (**C/C++**), Algorithms & Data Structures, Analog Signal Processing, Introduction to Computing, Introduction to Electronics, Linear Algebra with Computational Applications (**Python**), Discrete Structures, Physics: Electricity and Magnetism, Physics: Mechanics

Awards: Grainger Engineering Visionary Scholarship, Schaumburg Business Association Scholarship

Certifications: Anthropic: Claude Developer Platform 101, Intro to Model Context Protocol, Claude Code in Action

Involvements: Hoeft T&M Program, Phi Gamma Nu Professional Business Fraternity, Illinois Space Society, Illini Triathlon Club

TECHNICAL SKILLS

Programming Languages: C/C++, Python, TypeScript, SystemVerilog, Java, SQL, ReactNative, JavaScript, Assembly

Frameworks/Tools: Microsoft Azure, Git, Claude Code, Linux, CI/CD, Snowflake, Oscilloscope, Vivado, FPGA, Office 365

WORK EXPERIENCE

Nerdio Skokie, IL
Software Engineer Intern May 2026 – August 2026

- Architected a **TypeScript MCP server** integrating Microsoft Graph and Azure ARM to audit privileged Entra ID / Azure RBAC access across a **60+-tenant multi-cloud identity platform**; 12 tools, OAuth app-only auth, and Azure Container Apps hosting with Entra-authenticated HTTP transport
- Built an authenticated **LLM/API proxy** so internal apps can call Claude without exposing the Anthropic API key; validates Entra ID JWTs, injects a server-side key, proxies the Anthropic API, and enforces per-user allowlists, model allowlists, and rate limits; deployed to Azure App Service with Key Vault secrets and structured audit logging. **Servicing 5+ internal tools**
- Engineered a TypeScript **Azure Function App** multi-destination **webhook delivery platform** that replaced scattered automations with a queue-backed routing system to Teams, Azure Communication Services Email, or HTTPS webhooks, with an append-only Table Storage audit log; Entra Easy Auth and a custom RBAC admin dashboard on Static Web Apps, with GitHub Actions OIDC CI/CD

Solutions Engineer Intern May 2025 – August 2025

- Built and implemented multiple automation workflows, including a real time automated reply system for the company help forum and an email notification system for incoming support tickets; **reduced avg ticket response time by over 15%**
- Collaborated in a 4-person team using 1-2 week JIRA sprints and weekly standups to ship new internal tools
- Created autonomous SQL and Python processes to clean raw ticket data and deliver formatted reports to the CTO
- Developed Streamlit dashboards and web apps to visualize product usage, customer interaction, and web traffic

PERSONAL PROJECTS

First10 | Garmin Connect IQ, Monkey C, Embedded Programming, Constrained Hardware

- Built a Garmin wearable app that keeps an alarm active until a multi-sensor Awake Score (steps, sustained motion, and heart rate rise above a resting baseline) is passed, with sensors and alerts gated to the alarm phase
- Designed around Connect IQ limits: foreground-only vibration/tone and high-power CPU while open, keeping the AMOLED screen off and sensors idle during sleep for battery optimization, then enabling HR and accelerometer sampling when alarm fires
- Implemented difficulty presets, an HR-gated Hard mode resistant to wrist shakes, snooze limits, escalating alerts, and safety dismiss; added unit tests for scoring, hold timing, and degraded mode.

Portfolio Site | TypeScript, Next.js, Tailwind, Vercel

- Built colemurray.dev from scratch in Next.js and TypeScript, deployed on Vercel; pulls live stats from GitHub API

Simple Microprocessor | SystemVerilog, Xlinix Vivado, Spartan-7 FPGA

- Designed and built a 16-bit CPU in SystemVerilog on a Spartan-7 FPGA, with a multi-cycle FSM controller, register file, ALU, and block-RAM memory interface. Verified in simulation and on hardware with self-written testbenches and on-board test programs, tracing fetch and execute states through waveforms to debug control signal and timing bugs.